25901116 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCBOT200 - ARCHEGONIATES

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer

Answer any **10** questions

Each question carries **1** marks

1. Which generation produces spores in archegoniates ? [K] / [CO1]
2. Name one type of cell seen in pericycle of Pinus needles. [K] / [CO3]
3. What is fragmentation? [U] / [CO1]
4. What is dictyostele? [K] / [CO3]
5. What is homospory? [K] / [CO3]
6. Which pteridophyte is used as a biofertilizer in paddy fields? [U] / [CO4]
7. What is peat moss? [K] / [CO4]
8. In which year was the modern classification of gymnosperms proposed by Christenhusz et al.? [K] / [CO1]
9. State whether female gametophyte is dependent or non-dependent on Pinus sporophyte. [U] / [CO3]

10. Which plant group is the first vascular plant group to have appeared on Earth? [K] / [CO1]

11. What is the key characteristic of the gametophyte in Hepaticopsida? [K] / [CO1]

12. Name the male and female sex organs in the gametophyte of Psilotum. [K] / [CO3]

[1x10 = 10]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

13. Explain progressive dominance and independence of the sporophyte in archegoniates [U] / [CO1]

14. Illustrate the life cycle of bryophytes. [U] / [CO1]

15. Give a detailed account of the mature antheridium of Anthoceros. [U] / [CO3]

16. Give an account on PPG system of classification. [An] / [CO2]

17. Describe the structure of Pteris prothallus. [U] / [CO3]

18. Write short notes on ornamental gymnosperms with examples. [K] / [CO4]

19. Explain the anatomy of capsule in Pogonatum. [U] / [CO3]

20. Describe the structure and distinctive features of the megasporophyll of Cycas. [U] / [CO3]

[5x6 = 30]

Part C

Essay

Answer any **1** questions

Each question carries **10** marks

21. With the help of labelled diagram, explain the structure of strobilus in Selaginella. [U] / [CO3]

22. Describe the life cycle of *Cycas*, from sporophyte to gametophyte, highlighting reproductive adaptations. [U] / [CO3]

[10x1 = 10]